RESOLVEGRID AI

Autonomous Multi-Agent Civic Infrastructure Grievance Resolution Platform

OVERVIEW

ResolveGrid AI is an autonomous multi-agent public-infrastructure grievance resolution platform designed for municipal services. It brings citizens, administrators and field crews together through three dedicated portals, creating a transparent workflow for reporting, prioritising, assigning and tracking civic incidents.

THE PROBLEM

Municipal bodies receive complaints about potholes, water leakage, overflowing garbage, damaged streetlights and other public infrastructure issues. Traditional complaint processes can involve manual triage, delayed assignment, duplicate reports and limited visibility for citizens. Critical safety issues also need to be identified and escalated quickly.

OUR SOLUTION

ResolveGrid AI streamlines the grievance lifecycle—from citizen report to field-level action and evidence-based closure. Citizens can submit text or voice complaints and provide a location. The system analyses the report, checks for nearby duplicates, assesses urgency and routes work to an appropriate field crew. Administrators retain oversight through approval gates for critical hazards, while workers receive assignments, update progress and submit completion evidence.

CORE AI AGENTS & WORKFLOW

- **Incident Intelligence:** interprets complaint text and voice transcripts, extracts relevant details and classifies the incident.
- **Priority Assessment:** evaluates severity and flags public-safety risks for human approval.
- **Resource Allocation:** identifies suitable field crews using geographic proximity, department and workload.
- **Multi-Channel Coordination:** supports notifications and spoken work instructions for field personnel.
- **Evidence Verification:** evaluates submitted completion evidence and supports the resolution process.

A dedicated Duplicate Detection node checks nearby active incidents using a 250-metre geographic radius, helping reduce repeated dispatches.

GNANI.AI VOICE INTEGRATION

Gnani Prisma Speech-to-Text enables voice-based complaint submission in Marathi, Hindi and English. Gnani Timbre Text-to-Speech generates spoken instructions for field crews. Evon reasoning and Warp speech-to-speech are represented with fallback handling because the configured API permissions do not currently allow confirmed live access to those services.

PLATFORM & TECHNOLOGY

The platform includes a Citizen Portal for complaint submission and tracking, a Field Crew Hub for assignments and evidence uploads, and an Admin Operations Command for incident oversight, approvals, workforce management and AI activity monitoring. Technology stack: Python 3.13, FastAPI, React 18, TypeScript, Vite, Tailwind CSS, SQLite, LangGraph, WebSockets, JWT authentication and Gnani.ai APIs.

TESTING & EXPECTED IMPACT

The final project audit reported 49/49 backend tests, 12/12 Playwright tests and 25/25 local smoke-test steps passed. ResolveGrid AI aims to improve response coordination, prioritise public-safety incidents, reduce duplicate work and give citizens clearer visibility into the status of municipal complaints.